

Game Theory

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- Game theory has wide range of applications including psychology, evolutionary biology, war, politics, economics and business.

Examples

Example

1. Launching advertising campaigns for competing products.

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1. Launching advertising campaigns for competing products.
2. Planning strategies for war battles.

Basic terms in Game theory

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- **Information set:** The information available at a given point in the game.
- **Equilibrium:** The point in a game where both players have made their decisions and an outcome is reached.

Types of strategies

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- 1 **Pure strategy:** If a player knows exactly what the other player is going to do, a deterministic situation is obtained and objective function is to maximize the gain.
- 2 **Mixed strategy:** If a player is guessing as to which activity is to be selected by the other on any particular occasion, a probabilistic situation is obtained and objective function is to maximize the gain.

Types of game

Depending on the number of players:

- ① **Two person game:** There are two players.
- ② **N person game:** More than two players.

Types of game

Depending on the gain and loss:

- ① **Zero sum game:** The total gains and losses of all players sum to zero in all possible outcomes.

Example

Poker, chess, any sports

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Example

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2. Collaboration between two companies may benefit both.

Two person zero sum game

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- Associated with each pair of strategies, is the payoff one player receives from the other.
- Such a situation is known as two person zero sum game. (Because, a gain by one player is an equal loss by the other)

Payoff matrix

- Designing the two players as A and B with m and n strategies, respectively, the game is usually represented in terms of the payoff matrix to player A as,

$$\begin{array}{c} A_1 \\ A_2 \\ \vdots \\ A_m \end{array} \begin{bmatrix} B_1 & B_2 & \dots & B_n \\ a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

- This representation indicates that if A uses strategy i and B uses strategy j , the payoff to A is a_{ij} and payoff to B is $-a_{ij}$.

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- When these two values are equal, the corresponding strategies are called optimal strategies and the game is said to have saddle point.
- The value of the game is given by saddle point, represented as V .
Where $V = \underline{V} = \overline{V}$.

Saddle point

- It is a position in the payoff matrix where the maximum of the row minima coincides with the minimum of the column maxima.

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- It is a position in the payoff matrix where the maximum of the row minima coincides with the minimum of the column maxima.
- A game is said to be **fair** if
Maximin value = Minimax value
- The game is said to be **strictly determinable** if
Maximin value = Minimax value

Solution for pure strategy game

- Obtain payoff matrix.
- Find row minimum value of each row.
- Find the maximum value among all row minimum values (maximin, $\underline{V}$).
- Find column maximum value of each column.
- Find the minimum value among all column maximum values (minimax, $\overline{V}$).
- Since pure strategy is used, saddle point $V = \underline{V} = \overline{V}$.
- The optimum strategy corresponds to the entry (A_i, B_j) for which the value is V .

Example

1. Two companies, A and B , sell two brands of flu medicine. Company A advertises in radio (A_1), television (A_2), and newspaper (A_3). Company B , in addition to using radio (B_1), television (B_2), newspaper (B_3), also mails brochures (B_4). Depending on the effectiveness of each advertising campaign, one company can capture a portion of the market from the other. The following matrix summarizes the percentage of the market captured or lost by company A .

	B1	B2	B3	B4
A1	8	-2	9	-3
A2	6	5	6	8
A3	-2	4	-9	5

Example

2. Following is the payoff matrix in dollars for a two person zero sum game. Using minimax criterion, find the best strategy for each player.

	B1	B2	B3
A1	-3	5	-3
A2	1	3	5
A3	-3	-7	11

Example

3. Solve the game whose payoff matrix is given by

	B1	B2	B3
A1	1	3	1
A2	0	-4	-3
A3	1	5	-1

Example

4. Player A can choose his strategies from (A_1, A_2, A_3) only. While player B can choose from the set (B_1, B_2) only. The rules of the game state that the payments should be made in accordance with selection of strategies.

Strategy pair selected	Payment to be made
A_1B_1	Player A pays Rs. 1 to player B
A_1B_2	Player B pays Rs. 6 to player A
A_2B_1	Player B pays Rs. 2 to player A
A_2B_2	Player B pays Rs. 4 to player A
A_3B_1	Player A pays Rs. 2 to player B
A_3B_2	Player A pays Rs. 6 to player B

What strategy should A and B play in order to get the optimum benefits of the play?

Example

5. Find the range of values of P and Q which will render the entry $(2, 2)$ a saddle point for the game. (Ignore the values of P and Q to determine saddle point.)

		Player B		
Player A	1	2	4	5
	2	10	7	Q
	3	4	P	6

Example

5. For what values of γ , the game with following payoff matrix is strictly determinable. (Ignore the values of γ to determine saddle point)

		Player B		
		γ	6	2
Player A	-1	γ	γ	-7
	-2	-2	4	γ

Solution of mixed strategy games

- Games with mixed strategies can be solved either graphically or by linear programming.
- The graphical solution is suitable for games with no saddle point and having a payoff matrix of order $n \times 2$ or $2 \times n$.

Graphical solution of games

- 1 Start with $2 \times n$ games in which player A has two strategies A_1 and A_2 .
- 2 Player A mixes strategies A_1 and A_2 with probabilities x_1 and $1 - x_1$, $0 \leq x_1 \leq 1$.
- 3 Player B mixes strategies $B_1, B_2, \dots, B_n$ with probabilities $y_1, y_2, \dots, y_n$ respectively, $y_j \geq 0$, $1 \leq j \leq n$ and $y_1 + y_2 + \dots + y_n = 1$.
- 4 Payoff matrix for A is,

	B1	B2		B4
A1	a_{11}	a_{12}	$\dots$	a_{1n}
A2	a_{21}	a_{22}	$\dots$	a_{2n}

- 5 A 's expected payoff corresponding to B 's j^{th} pure strategy is

$$(a_{1j} - a_{2j})x_1 + a_{2j}, \quad j = 1, 2, \dots, n$$

- 6 Player A seeks the value of x_1 that maximizes the minimum expected payoffs. That is,

$$\max_{x_i} \left\{ \min_j \{ (a_{1j} - a_{2j})x_1 + a_{2j} \} \right\}$$

- ① Suppose if B has only two strategies and A has n strategies, then B 's expected payoff corresponding to A 's pure strategy is

$$(a_{j1} - a_{j2})y_1 + a_{j2}, j = 1, 2, \dots, n$$

- ② Player B seeks the value of x_1 that maximizes the minimum expected payoffs. That is,

$$\min_{y_i} \left\{ \max_j \{ (a_{j1} - a_{j2})y_1 + a_{j2} \} \right\}$$

Procedure to solve $2 \times n$ matrix

- 1 Draw two vertical axis 1 unit apart. The two lines are $x_1 = 0, x_1 = 1$.
- 2 Take the points of the first row in the payoff matrix on the vertical line $x_1 = 1$ and the points of the second row in the payoff matrix on the vertical line $x_1 = 0$.
- 3 The point a_{1j} on axis $x_1 = 1$ is then joined to the point a_{2j} on the axis $x_1 = 0$ to give a straight line. Draw ' n ' straight lines for $j = 1, 2, \dots, n$ and determine the highest point of the lower envelope obtained. This will be the **maximin point**.
- 4 The two or more lines passing through the maximin point determines the required 2×2 payoff matrix. This in turn gives the optimum solution by making use of analytical method.

Procedure to solve $n \times 2$ matrix

- 1 Draw two vertical axis 1 unit apart. The two lines are $x_1 = 0, x_1 = 1$.
- 2 Take the points of the first column in the payoff matrix on the vertical line $x_1 = 1$ and the points of the second column in the payoff matrix on the vertical line $x_1 = 0$.
- 3 The point a_{j1} on axis $x_1 = 1$ is then joined to the point a_{j2} on the axis $x_1 = 0$ to give a straight line. Draw ' n ' straight lines for $j = 1, 2 \dots n$ and determine the lowest point of the upper envelope obtained. This will be the **minimax point**.
- 4 The two or more lines passing through the minimax point determines the required 2×2 payoff matrix. This in turn gives the optimum solution by making use of analytical method.

Example

1. Use graphical method to reduce the following game and hence solve.

	B1	B2	B3
A1	3	-3	4
A2	-1	1	-3

Example

2. Solve the following using graphical method. Write the strategies adopted by each player and value of the game.

	B1	B2	B3
A1	6	4	3
A2	2	4	8

Example

3. Obtain the optimal strategies for both players and value of the game for two person zero sum game, whose payoff matrix is as follows.

	B1	B2
A1	1	-3
A2	3	5
A3	-1	6
A4	4	1
A5	2	2
A6	-5	0

For simplicity, the size of the payoff matrix can be reduced by using the principle of dominance.

Dominance rule

- If each element in a row is less than or equal to the corresponding element in another row, then that row is said to be dominated and can be removed from the matrix.
- If each element in a column is greater than or equal to the corresponding element in another column, then that column is said to be dominated and can be removed from the matrix.
- Dominance need not be based on the superiority of pure strategy only. A given strategy can be dominated if it is inferior to an average of two or more other pure strategies.

Example

1. Use dominance principle to solve the following game.

		Player B				
Player A	6	15	30	21	6	
	3	3	6	6	4	
	12	12	24	36	3	

Example

2. Solve the following game, use the dominance method to reduce the matrix. Write the strategies adopted by each player and value of the game.

		Player B			
Player A		3	2	4	0
		3	4	2	4
		4	2	4	0
		0	4	0	8